
title:... "Time-Off... Micro... ervice... -... Solution... Quality:... Non-Functional... A... ..e... ..ment"
author:... "ExampleHR... En... ineerin... "
date:... "2026-05-12"
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font... ize:... 11pt
mainfont:... DejaVu... Serif
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linkcolor:... blue

#... Executive... Summary

A... ..a... ..olution... architect... reviewin... .. thi...y... ..tem,... the... non-functional... ..po... ..ture... ..i...tron... in... **correctne... .. ,... reliability,... and... auditability...**
... the... three... area... .. that... matter... mo... ..t... for... a... financial-
...rade... leave... mana... ..ement... ..y... ..tem... There... are... deliberate... trade-off... in... **horizontal... calability...** (SQLite... ..ceilin... ..) and... **operational... ..ob... ..ervability...**

#... 1... Correctne... ..&... Data... Inte... rity... *****

This... ..i... .. the... hi... ..he... ..t-priority... NFR... for... a... ..balance-mana... ..ement... ..y... ..tem,... and... ..it... ..i... .. the...tron... ..e... ..t... area... of... the...olution.

What... wa...done... ..ri... ..ht:

-... **The... invariant... ..i...enforced... ..at... the... databa... ..e... layer,... not... ju... ..t... in... app**
lock... UPDATE... mean...no... ..application... ..bu...can...ilently... ..corrupt... ..a... ..balance... without.

-... **Two...uard...a... ..ain... ..t... over-allocation...** The... local...hadow... ..check... ..catche...
time... HCM... ..pre-fl...ht... ..catche...tale... ..hadow... ..ca... ..e...Even... ..if... the... HCM...ilently

-... **The... audit... ..lo...i...a... ..fir... ..t-cla...citizen...** Every... ..balance... ..mutation... .. (crea
only... ..no... UPDATE... or... DELETE... operation...exi... ..t... in... the... codeba... ..e.

-... **HCM... call... placement... ..i...correct...** The...etBalance... call... happen...before... ..the

Score:... 5... 5... The... core... invariant... available... =... hcm_balance... -... pendin... _day... ..
condition... ..te... ..t... that... a...ert...exactly... one...ucce...and... one... failure... for... two...
limit... reque... ..t...

#... 2... Reliability... &... Fault... Tolerance... *****

What... wa...done... ..ri... ..ht:

-... **HCM... unavailability... ..i...handled...racefully...** If... the... HCM... ..i...down... ..durin...
...no... ..partial...tate.

-... **Retry... with... backoff...** The... HCM... client... retrie...3... time...on... 5xx... timeout... (100

-... **Optimistic-lock...retry...** Balance... write... retry... up... to... 5... time... with... jitter... This...
 -... **PRAGMA...busy_timeout...=...5000...** SQLite... will... wait... up... to... 5... second... for... a...
 -... **Transactional...atomicity...** Every... multi-step... operation... (create... request... +... update...
 ... either... all... succeed... or... none... do.

Where... it... fall...hort... (one...tar...deducted):

-... **No...outbox...pattern...for...approval...failure...** If... the... HCM... deduct... call... succeed...

Score:...4...5

#...3...Scalability...*****

What... was... done...right:

-... **Stateless...service...design...** No... in-process... operation... state,... no... connection...
 local... cache... The... service... could... it... behind... a... load... balancer... with... multiple... in...
 ... the... only... constraint... i... the... database... e.

-... **SQLite...WAL...mode...** WAL... allow... concurrent... reader... even... during... write... F...
 heavy... workload... (employee... checkin... balance...),... this... i... nificantly... improve... th...

-... **Parallelized...API...** All... li... t... endpoint... enforce... a... maximum... of... 100... re... ult...

-... **Batch...sync...process...e...per-entry...** Each... balance... entry... in... a... batch... sync...
 row... batch... do... n... t... create... a... 10,000-row... monolithic... transaction... that... lock... the... D...

Where... it... fall...hort:

-... **SQLite...i...a...in...le-writer...bottleneck...** The... BEGIN... IMMEDIATE... pattern... en...
 process... databa... e... horizontal... write... calin... require... mi... ratin... to... Po... t... r...

-... **The...repository...pattern...abstract...the...DB...layer...**
 ... wappin... TypeORM... driver... from... better-...qlite3... to... p... require... chan...

Score:...3...5 -... Appropriate... for... the... a... i... nment... (SQLite... i... pecified),... but... ex...

#...4...Security...*****

What... was... done...right:

-... **HCM...sync...endpoint...u...e...timin...-...afe...key...compari...on...crypto.timin...**
 ... a... detail... many... implementation... mi... ..

-... **JWT...uard...i...in...place...** (... tub,... but... the... uard... interface... i... correct... fo...

-... **Input...validation...on...all...DTO...cla...-validator...with...whitelis...t:...true...**
 ... protection... a... ain... t... ma... -a... i... nment... attack...

-... **Rate...limitin...on...all...endpoint...** 100... request... t... minute... per... IP... via... @ne...

-... **No.....ecret.....in...code...** All.....en...itive...value.....(HCM...API...key,...JWT.....ecr
fa...t...validation...at.....tartup.

-... **Error...me.....a...e.....don...t...leak...internal.....** The.....lobal...exception...filter...

Where...it...fall.....hort:

-... **JWT...i.....a.....tub...** Real...authentication...with...role-
ba...ed...acce.....(employee...v.....mana...er)...i.....not...implemented...Employee.....could...

-... **No...HTTPS...enforcement...** TLS...termination...i.....a.....umed...to...happen...at...the...

Score:..4...5..- The.....ecurity...primitive.....are...correct.....the...JWT.....tub...i.....an.

#...5...Ob...ervability...*****

What...wa.....done...ri...ht:

-... **Reque...t...ID...propa...ation...** Every...reque...t.....enerate.....a...UUID,...attache.....it.

-... **Structured...lo.....in.....** Ne...tJS.....built-in...lo.....er...output.....JSON-
compatible.....structured...entrie.....with...context...label.....(**TimeOffService...**,...**HTTP...**,

-... **Bu...ine.....-level...audit...trail...balance_audit_lo.....** provide.....a...queryable...hi.
...more...valuable...for...op.....compliance...than...application...lo.....

-... **Di...crepancy...detection...** Batch.....ync...explicitly...detect.....and...lo.....ne...ative-
available.....cenario.....a.....**WARN**...with.....structured...field.....(employee,...location,...diff...amc

Where...it...fall.....hort:

-... **No...metric.....exporter...** There.....no...Prometheu.....endpoint,...no...Stat...D...inte...

-... **No...di...tributed...tracin.....** The...**X-Request-Id**...i.....et...but...not...propa...ated.

Score:..3...5..- Sufficient...for...a...take-home.....need.....Prometheu.....+...OpenTelemetry...

#...6...Maintainability...*****

What...wa.....done...ri...ht:

-... **Sin...le...re...pon...ibility...throu...hout...** **HcmClientService**...i.....the...only...cla.....
lock...mutation.....**TimeOffService**...own.....the.....tate...machine...Each...concern...ha.....exa

-... **The...repo...itory...pattern...** Bu...ine.....lo...ic...never...write.....raw...SQL...The...re

-... **Explicit...invariant.....documented...** **CLAUDE.md**...and...code...comment.....explain...*why*...
only...Future...maintainer.....won...t...accidentally...“fix”...the...e.

-... **92%...te...t...covera...e...** The...105-te...t.....uite...provide.....a...re...re.....ion...net...t

-... **Swa.....er.....OpenAPI...documentation...** Every...endpoint,...DTO...field,...and...error
...and...the.....pec.....tay.....in.....ync...with...the...code...automatically...(no.....eparate...do

Score:...5...5

#...Overall...A.....e.....ment

...NFR...	...Score...	...Note.....
...Correctne.....&...Data...Inte...rity...	*****	...Core...invariant...i.....provably...maintained...
...Reliability...&...Fault...Tolerance...	*****	...Outbox.....ap...i.....the...only...notable...ri...k...
...Scalability...	*****	...SQLite...ceilin.....mi...ratin.....to...PG...i...
...Security...	*****	...JWT.....tub.....all...other...primitive.....are... ready...
...Ob...ervability...	*****	...No...metric.....tracin.....audit...lo.....i.....
...Maintainability...	*****	...Clean.....eparation,...documented...invariant...,...9
...Overall...	*****	...Production- ready...with...three...known...up...rade...path.....

The...three...production...up...rade...path...,...in...priority...order:

- 1...Swap...SQLite...>...Po...t...reSQL...(1...confi.....chan...e...in...Databa...eModule.....r
- 2...Add...Prometheu.....metric.....endpoint...(1...new...Ne...tJS...module.....no...domain.
- 3...Implement...JWT...verification...+...RBAC...(replace...the.....uard.....tub.....no.....0